

Hetang Mehta

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EDUCATION

University of Alberta

BSc Computer Engineering (Co-op Program)

April 2027

Edmonton, AB

EXPERIENCE

Machine Learning Engineer

Alberta BIONIX (Student Club)

Sep 2025 – Present

Edmonton, AB

- Processed and analyzed a **26GB EMG gait dataset** using **Python (Pandas, NumPy, Scikit-learn)**, building feature engineering pipelines to extract biomechanical signals used for prosthetic knee control.
- Developed and evaluated **machine learning models (Random Forest, SVM)** to classify prosthetic trial participant viability from EMG activation patterns.
- Designed and trained **CNN models for real-time knee flexion prediction**, achieving **87% accuracy** across locomotion modes (sitting, standing, walking, running).

Systems Developer Student

Canadian Natural Resources Limited (CNRL)

Jan 2024 – Aug 2025, May 2026 – Present

Calgary, AB

- Improved productivity by **30%** for **500+** engineers by owning the design and delivery of a PWA built with **C#, .NET, Blazor, REST APIs**, and **Azure CI/CD**.
- Reduced defects by **33%** and saved **\$40K** annually by redesigning **Oracle SQL** schemas, optimizing queries, and enforcing data integrity constraints.
- Reduced test latency from **10 days to 4 minutes**, saving **\$60K** by building **300+** automated unit tests and validation pipelines in **C#** and **VB.NET**.

Data Analyst Intern

Suncor Energy

Sep 2023 – Dec 2023

Fort McMurray, AB

- Developed **3 Python-based automation tools** for SAP workflows (data maintenance, updates, manipulation), saving **30+ hours/week** of manual work and **\$26K** annually for planners and schedulers.
- Cleaned and processed large datasets using **Python, Excel, and VB** to ensure data quality, then built analytical dashboards in **Power BI** to support data-driven decisions across the team.

PROJECTS

Batch Normalization Reproduction | *Python, PyTorch, NumPy, Matplotlib*

- Reproduced Ioffe & Szegedy's ICML 2015 **Batch Normalization** paper by implementing CNNs **with and without BatchNorm layers in PyTorch**, then benchmarking convergence and accuracy on **MNIST and CIFAR-10** to validate the paper's core claims.

Multithreaded Concurrent Server | *C, POSIX Threads, Sockets*

- Built a **multithreaded TCP server in C** handling concurrent client read/write requests with data consistency guarantees via **mutexes and reader-writer locks**, using a thread-pool architecture to manage connection load and prevent race conditions.

ATS Resume Analyzer | *Demo | Next.js, TypeScript, React, Google Gemini AI*

- Built a **GenAI-powered** full-stack web app with **Next.js 15, React, and TypeScript** that uses **Google Gemini LLMs** to score resume-to-job-description compatibility and surface keyword and optimization recommendations.

Boxed | *React, TypeScript, Node.js, Express, Auth0, Cloudflare R2, Docker*

- Built a cloud storage platform with secure uploads, retrieval, and **role-based access via Auth0 (OAuth 2.0)**, exposed through **RESTful APIs** and deployed on **Docker** with **S3-compatible storage (Cloudflare R2)**.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, SQL, Go, Swift, Kotlin, VB.NET, Ruby

ML / AI: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Gemini API

Cloud / DevOps: AWS, Azure (CI/CD), Cloudflare R2, Docker, Kubernetes, Git, GitHub, Linux

Web / Frameworks / Tools: React, Next.js, Node.js, Express, Spring Boot, .NET, Blazor, Flask, RESTful APIs, Auth0

Databases: SQL (Oracle, PostgreSQL, MySQL), NoSQL (MongoDB)